

AMOD5430 Project

Hamza Chaudhry

2025-07-24

```
library(ggplot2)

setwd("~/Desktop/AMOD5430")

# Netflix dataset im using
netflix <- read.csv("netflix_titles.csv", stringsAsFactors = FALSE)

# View column names and sample data
head(netflix)
```

```
## show_id type title director
## 1 s1 Movie Dick Johnson Is Dead Kirsten Johnson
## 2 s2 TV Show Blood & Water
## 3 s3 TV Show Ganglands Julien Leclercq
## 4 s4 TV Show Jailbirds New Orleans
## 5 s5 TV Show Kota Factory
## 6 s6 TV Show Midnight Mass Mike Flanagan
```

```
##
## 1
## 2 Ama Qamata, Khosi Ngema, Gail Mablane, Thabang Molaba, Dillon Windvogel, Natasha Thahane, Arno Gro
## 3
## 4
## 5
## 6 Kate Siegel, Zach Gilford, M
```

```
## country date_added release_year rating duration
## 1 United States September 25, 2021 2020 PG-13 90 min
## 2 South Africa September 24, 2021 2021 TV-MA 2 Seasons
## 3 September 24, 2021 2021 TV-MA 1 Season
## 4 September 24, 2021 2021 TV-MA 1 Season
## 5 India September 24, 2021 2021 TV-MA 2 Seasons
## 6 September 24, 2021 2021 TV-MA 1 Season
```

```
## listed_in
## 1 Documentaries
## 2 International TV Shows, TV Dramas, TV Mysteries
## 3 Crime TV Shows, International TV Shows, TV Action & Adventure
## 4 Docuseries, Reality TV
## 5 International TV Shows, Romantic TV Shows, TV Comedies
## 6 TV Dramas, TV Horror, TV Mysteries
```

```
##
## 1 As her father nears the end of his life, filmmaker Kirsten Johnson stages his death in inventive an
## 2 After crossing paths at a party, a Cape Town teen sets out to prove whether a private-school s
## 3 To protect his family from a powerful drug lord, skilled thief Mehdi and his expert team of
```

```
## 4 Feuds, flirtations and toilet talk go down among the incarcerated women at the Orleans Justice Center in New Orleans.
## 5 In a city of coaching centers known to train India's finest collegiate minds, an earnest but unexcused student is expelled.
## 6 The arrival of a charismatic young priest brings glorious miracles, ominous mysteries and renewed faith in the power of God.
```

```
str(netflix)
```

```
## 'data.frame': 8807 obs. of 12 variables:
## $ show_id : chr "s1" "s2" "s3" "s4" ...
## $ type : chr "Movie" "TV Show" "TV Show" "TV Show" ...
## $ title : chr "Dick Johnson Is Dead" "Blood & Water" "Ganglands" "Jailbirds New Orleans" ...
## $ director : chr "Kirsten Johnson" "" "Julien Leclercq" "" ...
## $ cast : chr "" "Ama Qamata, Khosi Ngema, Gail Mababane, Thabang Molaba, Dillon Windvogel, ...
## $ country : chr "United States" "South Africa" "" "" ...
## $ date_added : chr "September 25, 2021" "September 24, 2021" "September 24, 2021" "September 24, 2021" ...
## $ release_year: int 2020 2021 2021 2021 2021 2021 2021 2021 1993 2021 2021 ...
## $ rating : chr "PG-13" "TV-MA" "TV-MA" "TV-MA" ...
## $ duration : chr "90 min" "2 Seasons" "1 Season" "1 Season" ...
## $ listed_in : chr "Documentaries" "International TV Shows, TV Dramas, TV Mysteries" "Crime TV Shows" ...
## $ description : chr "As her father nears the end of his life, filmmaker Kirsten Johnson stages his posthumous journey through dreams and memories to reclaim his legacy, highlight his family's role in the global HIV/AIDS epidemic."
```

Data Cleaning and Preparation

```
# Using the date format
netflix$date_added <- as.Date(netflix$date_added, format = "%B %d, %Y")

netflix$year_added <- format(netflix$date_added, "%Y")

# Cleaning the country and listed_in columns
netflix$country[is.na(netflix$country)] <- "Unknown"
netflix$listed_in <- as.factor(netflix$listed_in)
netflix$type <- as.factor(netflix$type)
```

Figure 1: The Number of titles added to Netflix overtime

```
ggplot(netflix, aes(x = year_added)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Figure 1: Number of Titles Added by Year",
       x = "Year Added", y = "Count of Titles") +
  theme_minimal()
```

Figure 1: Number of Titles Added by Year

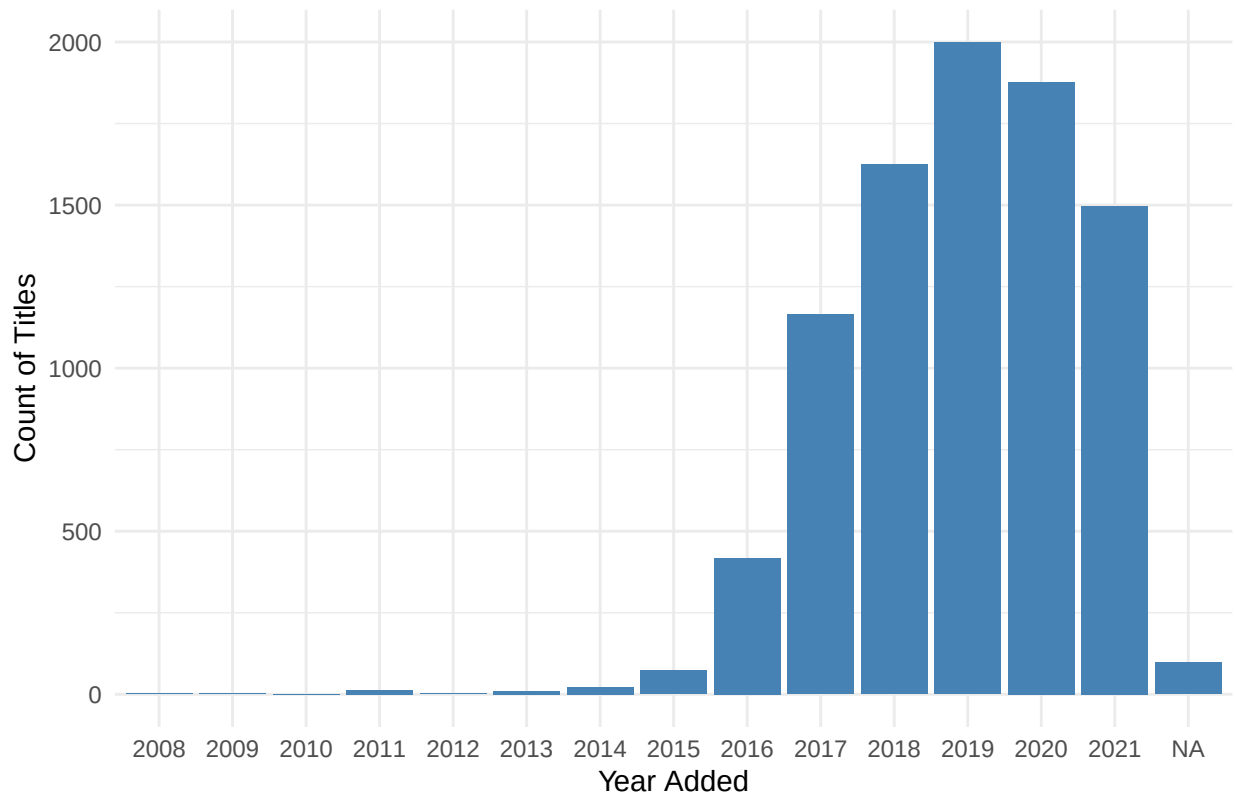


Figure 1 shows the number of titles added to Netflix each year based on the date_added column. # There's a clear surge from 2016 to 2020, suggesting Netflix aggressively expanded its library during this period. # The decline post-2020 may reflect strategic shifts or disruptions during the pandemic. # A small number of titles have missing dates, shown as "NA."

Figure 2: Movies vs TV shows on Netflix

```
ggplot(netflix, aes(x = type, fill = type)) +  
  geom_bar() +  
  labs(title = "Figure 2: Distribution of Content Type",  
        x = "Type", y = "Count") +  
  theme_minimal()
```

Figure 2: Distribution of Content Type

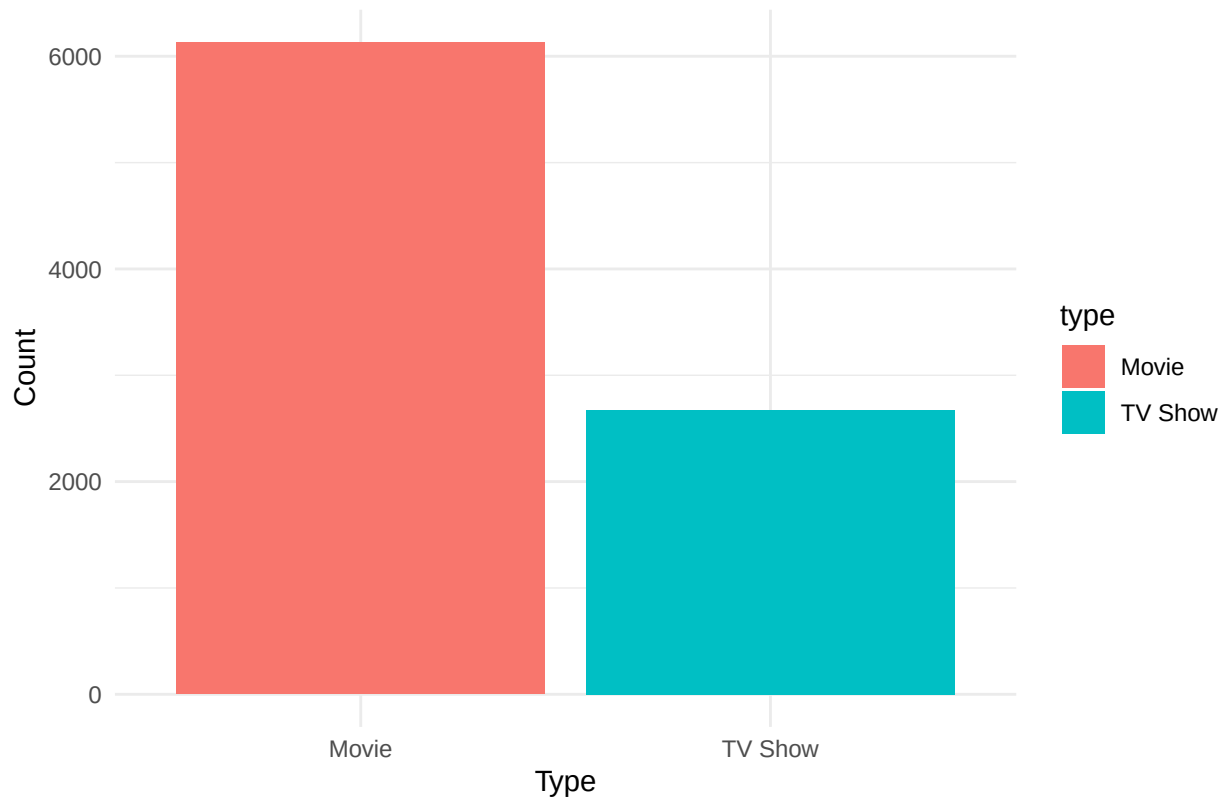


Figure 2 illustrates the overall distribution of content types on Netflix. # Movies dominate the platform, with more than twice the count of TV Shows. # This shows Netflix's preference for feature-length content, but serialized content is also important for user retention

Figure 3: Top 10 countries producing content on Netflix

```
top_countries <- sort(table(netflix$country), decreasing = TRUE)[1:10]
barplot(top_countries,
        col = "darkred",
        main = "Figure 3: Top 10 Countries by Number of Titles",
        ylab = "Count of Titles",
        las = 2)
```

Figure 3: Top 10 Countries by Number of Titles

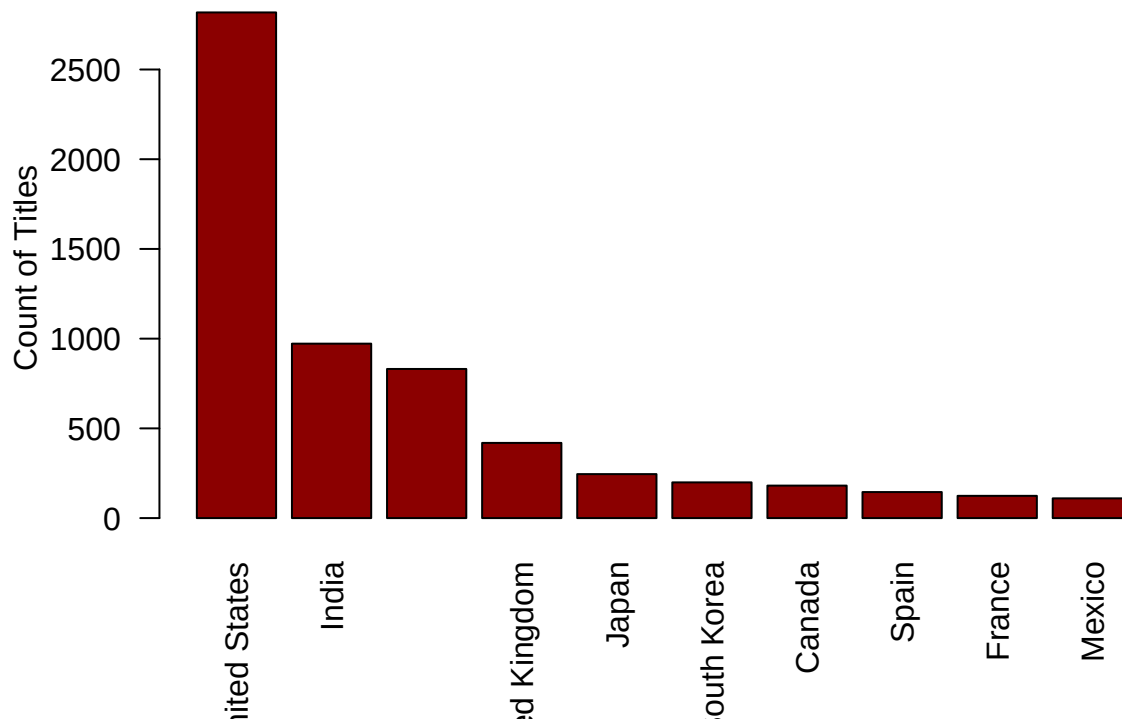


Figure 3 shows the top 10 countries producing content on Netflix. # The United States leads by a wide margin, followed by India and the UK. # This indicates that while Netflix has global reach, much of its content still originates from English-speaking or major media markets.

Figure 4: Top genres on Netflix

```
genre_counts <- table(netflix$listed_in)
top_genres <- sort(genre_counts, decreasing = TRUE)[1:10]

top_genres_df <- as.data.frame(top_genres)
colnames(top_genres_df) <- c("Genre", "Count")

library(ggplot2)
ggplot(top_genres_df, aes(x = reorder(Genre, Count), y = Count)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  coord_flip() +
  labs(title = "Figure 4: Top 10 Genres",
       x = "Genre", y = "Number of Titles") +
  theme_minimal() +
  theme(axis.text.y = element_text(size = 10))
```

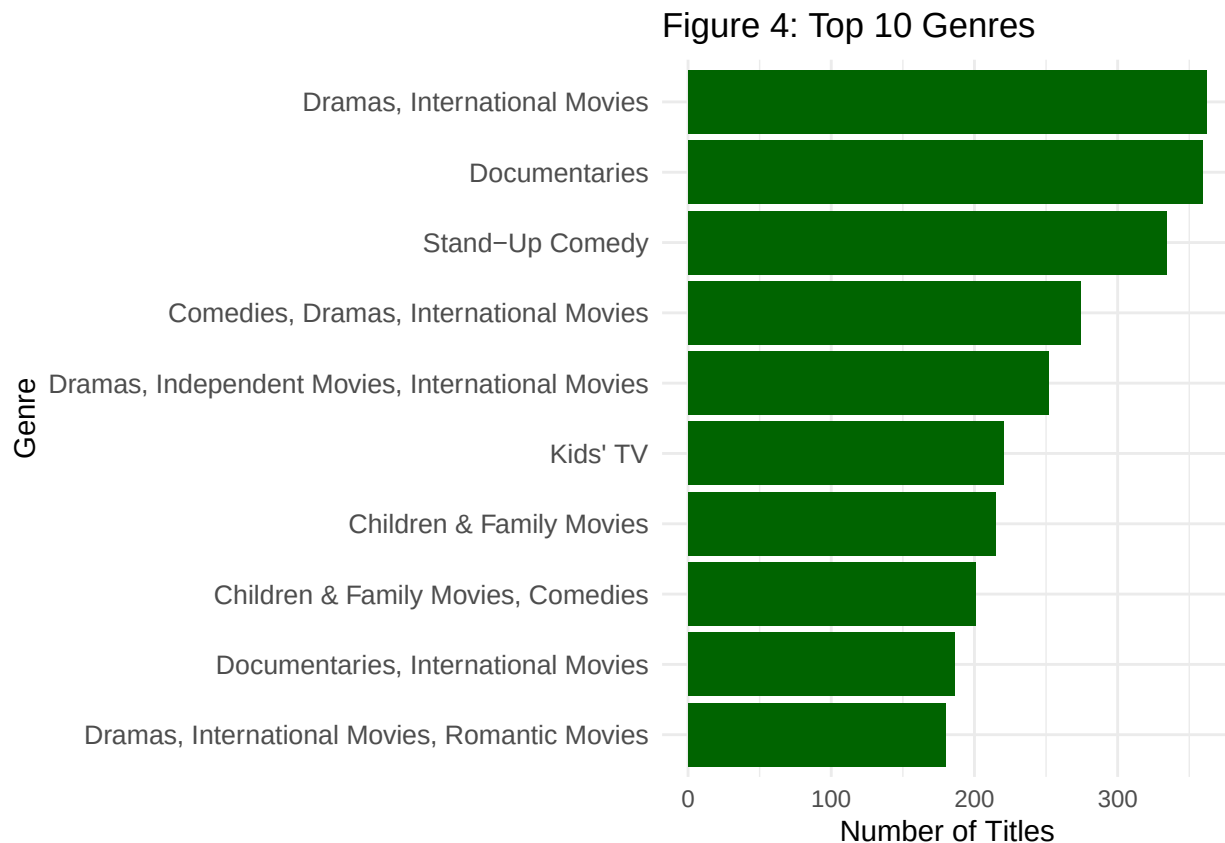


Figure 4 highlights the most common genres on Netflix. # International movies, documentaries, and stand-up comedy dominate the list. # The presence of multiple types of “International” genres shows Netflix’s global reach # Family-friendly and kid-oriented content also reinforce the previous point.

Figure 5: Movie durations on Netflix

```
movie_data <- subset(netflix, type == "Movie")
movie_data$duration_mins <- as.numeric(gsub(" min", "", movie_data$duration))

hist(movie_data$duration_mins,
     main = "Figure 5: Movie Duration Distribution",
     xlab = "Duration (minutes)",
     col = "skyblue",
     breaks = 30)
```

Figure 5: Movie Duration Distribution

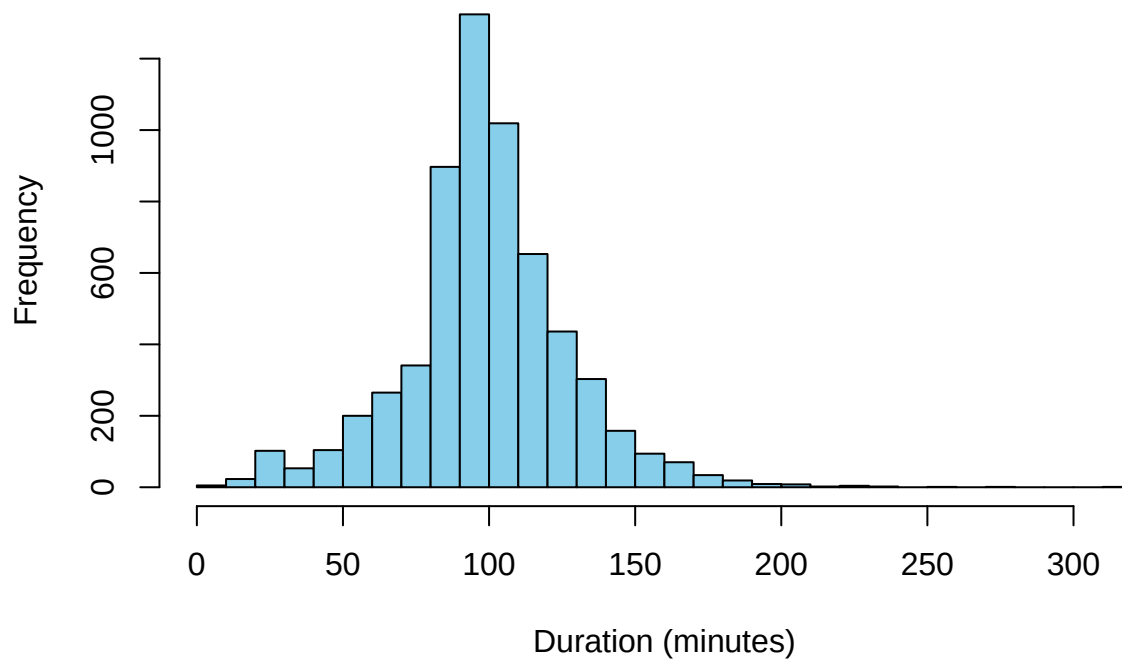


Figure 5 presents the distribution of movie durations. # Most films on Netflix fall in the 80–110 minute range, which is standard feature film length. # Very short and very long movies are rare, which is as expected.

— Conclusion —

Through our analysis, we found several trends in Netflix content.

- The number of titles added grew steadily, peaking in recent years.
- Movies dominate the platform compared to TV Shows.
- The United States leads in content production, but Netflix hosts titles from many countries.
- Drama and international genres are among the most common.
- Movie durations cluster around 90-100 minutes, with a few outliers.

These insights are important for content creators to understand Netflix's strategy and also to see what works and what does not.